

State of AI in Retail and Consumer Package Goods: 2026 Trends



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Survey Overview

AI in Retail and CPG: Key Trends for Industry Leaders

The retail and consumer packaged goods industries have reached a critical inflection point in their artificial intelligence journey. AI is being deployed throughout every line of business, from the back office to the supply chain, in stores and for digital commerce. Most importantly, those AI solutions are having a tangible, measurable impact on the bottom line, helping to streamline operations and boost employee productivity.

As experimentation gives way to widespread deployment, the competitive question has shifted from whether to invest in AI to how to design flexible, cost-efficient AI stacks that align with each organization's data, talent, and risk strategies. This increasingly means making deliberate choices about how to optimize inference at scale, how to balance open-source and proprietary models, and where to introduce new capabilities such as agentic AI and physical AI.

This third annual *NVIDIA State of AI in Retail and CPG* report examines the state of AI deployment across retail and CPG, analyzing business impact and investment trends, the evolution of open and flexible AI stacks, the increased importance of agentic AI, strategic use cases, and the critical implementation challenges around talent, data maturity, and inference optimization. It explores how organizations are navigating supply chain complexity with AI-driven solutions and concludes with forward-looking implications for competitive positioning in an AI-enabled industry.

Executive Summary

The role of AI in retail and CPG is expanding from isolated solutions to a comprehensive, enterprise-wide strategy. Today, AI supports marketing and advertising, optimizes supply chains, and enhances in-store experiences. This report highlights the key areas where AI is reshaping the industry, evaluating both the level of maturity and investment and spotlighting the top use cases that are driving tangible business outcomes.

Here are some of the top highlights from this year's report:

The AI Maturation Curve

58%

said that their organizations are actively deploying AI solutions, up from 42% in 2024.

33%

are currently assessing AI solutions, indicating a stabilization in evaluation activity following last year's broader exploratory phase at 47%.

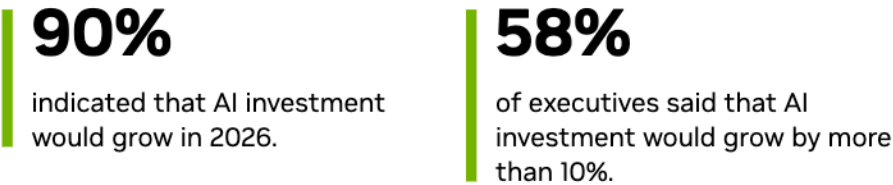
Across the business landscape, companies are transitioning from running numerous AI pilots to selecting and scaling solutions with proven business value. This trend is especially pronounced in retail and CPG, where active AI usage has significantly increased, and the share of organizations assessing AI has fallen 14 percentage points year over year.

How AI Is Impacting the Bottom Line



AI drives measurable impact in three key areas: operational efficiency, customer experience, and employee productivity. These statistics highlight AI’s dual impact on the bottom line: Not only is it a powerful driver of revenue growth, it also delivers substantial cost savings.

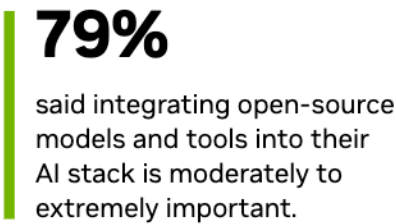
AI Success Leads to Growing Investment



As AI solutions demonstrate tangible business impact, nearly all respondents expect investment to rise or at least remain steady. Increased spending will enhance AI performance, uncover new use cases, and equip employees to integrate AI into their daily workflows.

Open-Source AI Gains Strategic Importance

As organizations scale AI, leaders are turning to open-source models and tools to adapt systems to their own data, retain flexibility in how they deploy, and tap into a fast-moving innovation ecosystem.



Agentic AI Is on the Rise

47%

said that their organizations are using or assessing AI agents.

20%

said AI agents are already actively deployed in their organizations.

Agentic AI—advanced systems that autonomously reason, plan, and execute complex tasks—is still early in enterprise adoption, yet retail and CPG organizations are already experimenting with it. AI agents are being used to accelerate operations, enhance customer experiences, and improve decision-making with real-time data.

Streamlining the Supply Chain

91%

said that AI is helping to decrease annual supply chain costs.

51%

said they're using AI to address operational throughput and efficiency in the supply chain.

The modern supply chain is extraordinarily complex, and retail and CPG companies are leveraging AI to streamline operations. The next frontier is deploying physical AI solutions like digital twins and robotics, and early implementations are already underway.

An In-Depth Look at the Results

Business Impact and Investment Trends

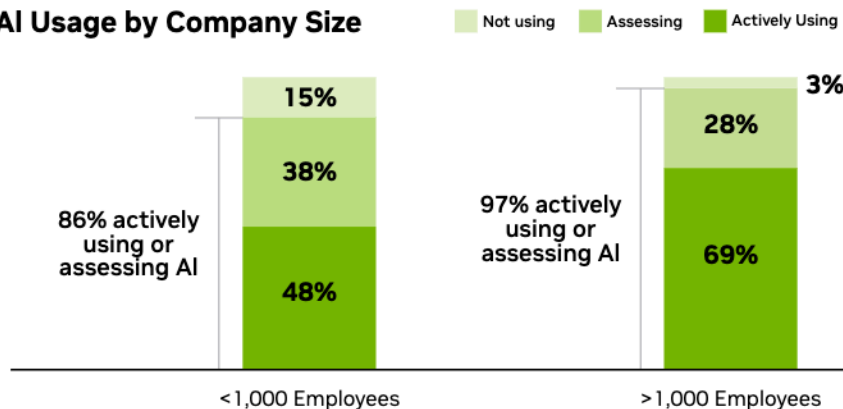
Ninety-one percent of retail and CPG organizations are now engaged with AI, signaling near-universal industry adoption. This momentum includes a sharp rise in active usage: 58 percent of respondents report that their companies are currently using AI, marking a 16-point increase from last year.

Larger organizations—those with more than 1,000 employees—lead with 69 percent actively using AI, compared to 48 percent among smaller firms. As experimentation gives way to widespread deployment, the competitive question has shifted from whether to invest in AI to how effectively organizations deploy it compared with peers.

91%

of the industry is engaged with AI.

AI Usage by Company Size



Proving the Business Case

AI has moved beyond aspirational technology to become a proven driver of business. The financial results validate the industry's aggressive investment posture: Revenue gains are widespread with 89 percent reporting positive impact and substantial with 57 percent achieving 5 percent or greater growth. At the same time, AI is helping decrease annual costs, according to 95 percent of respondents, with 58 percent reducing costs by 5 percent or more.

These dual impacts—simultaneous revenue growth and cost reduction—explain the executive commitment. AI is delivering the combination of top-line acceleration and bottom-line improvement, a value proposition that justifies continued investment even in uncertain economic environments.

89%

reported that AI has had a positive impact on revenue.

95%

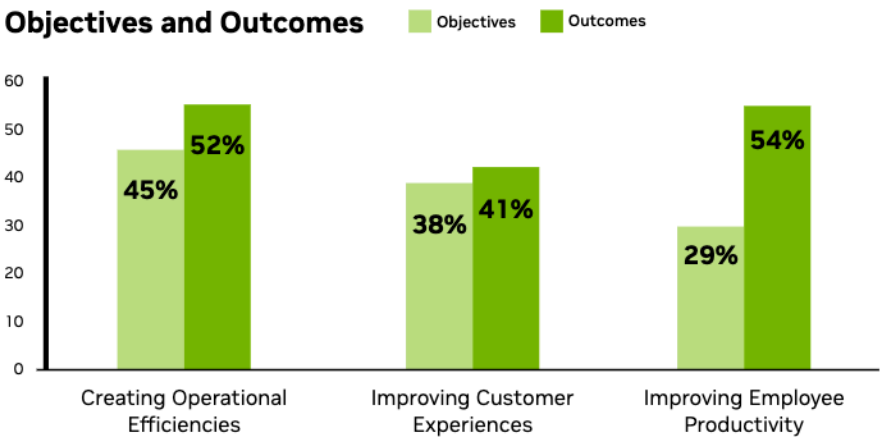
said that AI is decreasing annual costs.

Strategic Goals and Realized Outcomes

Survey respondents said that their organizations entered their AI journeys with three primary objectives:

- 1. **Creating operational efficiencies: 45 percent**
- 2. **Improving customer experience: 38 percent**
- 3. **Improving employee productivity: 29 percent**

Organizations are not only meeting but surpassing their initial objectives with AI. Fifty-four percent of respondents said AI has increased productivity, 52 percent reported improved operational efficiency, and 41 percent noted better customer experiences.



Investment Conviction

AI is shaping a promising future for retail and CPG, as intelligent tools across various business functions complement each other to streamline operations. Unsurprisingly, executive interest in AI investment is strong, with 92 percent saying that their AI budgets will increase in the next year, including 58 percent expecting boosts of more than 10 percent.

Organizations are doubling down on AI investments to drive real impact. Optimizing AI workflows and production tops the investment priority list at 36 percent, followed by identifying new use cases at 33 percent and training existing staff at 32 percent.

Beyond spending more, companies are deploying AI more strategically, creating a virtuous cycle of investment, learning, and performance gains.

92%
of executives plan to increase their AI budgets in the next year.

A Move Toward Open Source

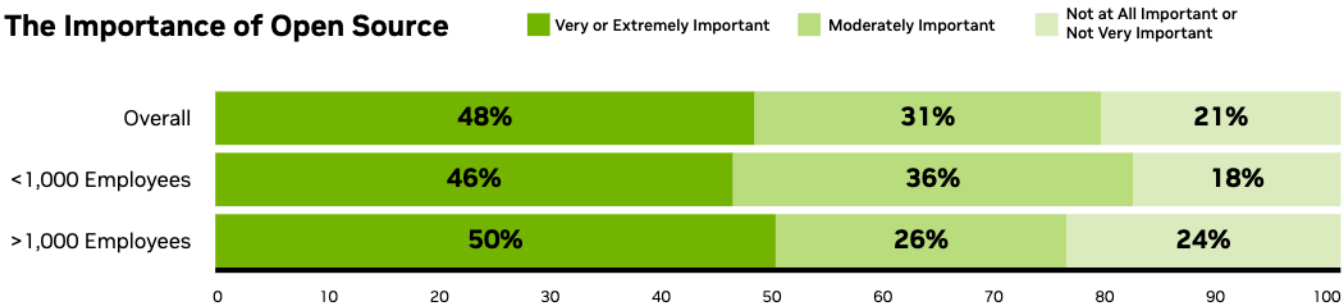
As generative AI matures, organizations are shifting from generalized tools to specialized solutions tailored to specific business needs. Open-source models and tools provide the flexibility to fine-tune AI with proprietary data while leveraging community innovations to accelerate development.

Retail and CPG organizations are embracing open source at rates comparable to other industries. Nearly half, 48 percent, consider it very to extremely important to integrate open-source models and tools into their technology stacks, while another 31 percent see it as moderately important.

48%

stated that open source is very to extremely important.

The Importance of Open Source



The Rise of Agentic AI: The Next Competitive Frontier

New to this year's survey are questions on agentic AI, systems that unlock new business value by autonomously reasoning, planning, and executing complex tasks.

Despite being an emerging technology, agentic AI is already entering enterprise workflows. In retail and CPG, 47 percent of respondents are using or assessing AI agents, including 20 percent actively deploying them and another 21 percent planning deployment within the next year.

47%

said they're already using or assessing agentic AI.

Strategic Objectives and Business Value

Organizations are deploying AI agents to address three interconnected business imperatives that traditional automation can't fully solve:

1. Increasing process speed and efficiency: 57 percent

Agentic AI can execute workflows continuously, handle exceptions autonomously, and scale instantly, transforming processes that once took hours or days into real-time operations.

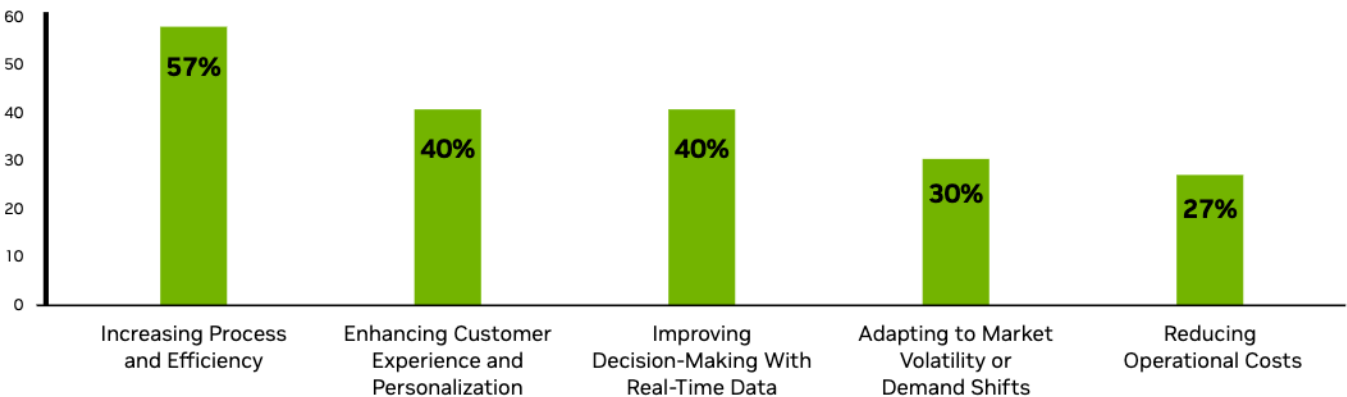
2. Enhancing customer experiences and personalization: 40 percent

Agentic AI can hold multi-turn conversations, understand nuanced customer needs, access information across systems, and take actions to resolve issues or complete transactions, delivering the responsiveness and personalization customers expect without increasing human support costs.

3. Improving decision-making with real-time data: 40 percent

AI agents can continuously monitor data streams, detect patterns needing intervention, evaluate options against business rules, and act or escalate as needed, enabling organizations to operate at the speed of their data rather than human review cycles.

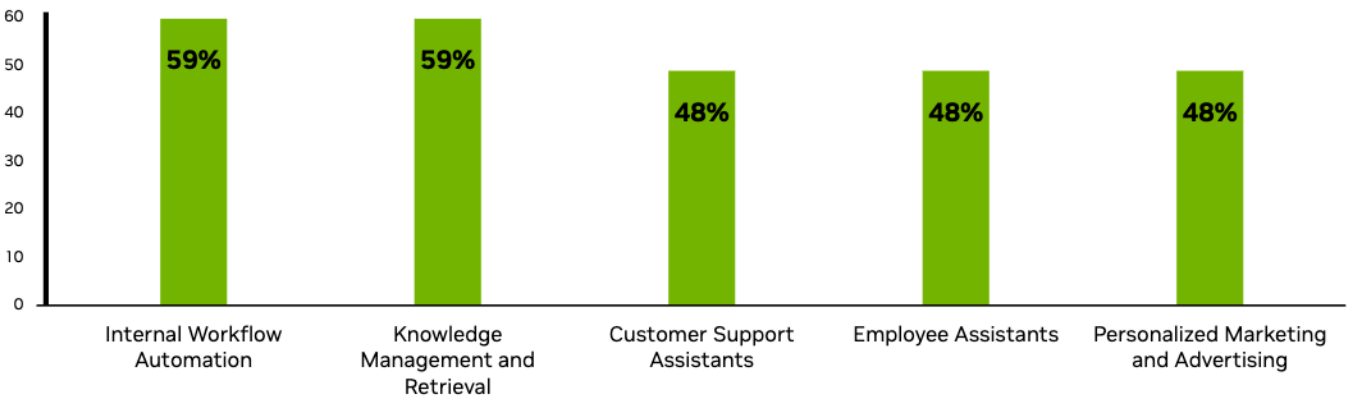
Top 5 Business Objectives Being Addressed With AI Agents



Agentic AI Implementation Areas

The deployment of agentic AI concentrates on use cases that combine high business impact with technical feasibility.

Top 5 Current Use of AI Agents



Internal operations are the top priority, with 59 percent citing both workflow automation and knowledge management and retrieval. These tackle a longstanding challenge: critical expertise trapped in documents, systems, and individuals. AI assistants that retrieve, synthesize, and apply institutional knowledge enable employees to leverage the organization’s collective intelligence.

Employee and customer support rank equally at 48 percent. Employee assistants boost productivity across functions, while customer support assistants handle routine inquiries, freeing humans to focus on complex cases. This balance highlights how AI delivers value both internally and externally.

Customer engagement applications include personalized marketing and advertising at 48 percent and shopping assistants at 41 percent. These agents go beyond analytics. They act on insights in real time, adjust messages, recommend products, and guide purchase decisions based on individual customer contexts.

Concern about AI agents using relevantly trained models shapes implementation strategies, according to 48 percent of respondents. Retail and CPG organizations operate with unique products, processes, customer behaviors, and business rules that general models may not understand. Companies are navigating the challenge of tailoring foundation models with proprietary data and domain expertise, increasingly turning to open-source models and tools to maintain performance while avoiding long-term technical debt.

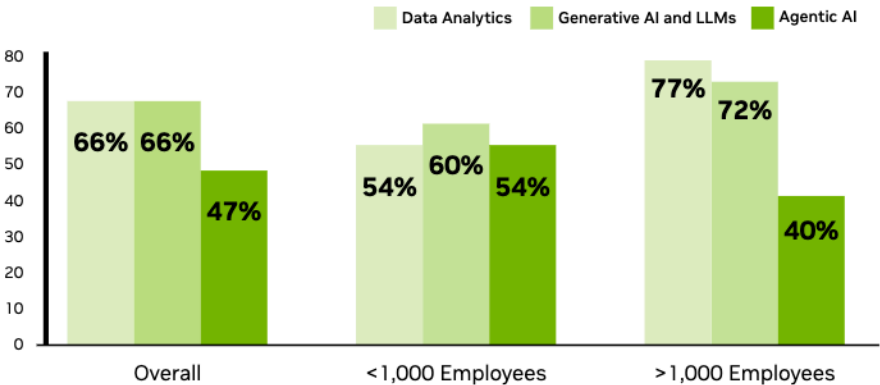
How Retail and CPG Are Leveraging AI Across the Value Chain

Organizations are increasingly deploying AI technologies to drive value across their entire business. Generative AI, large language models, and advanced data analytics are leading the way, with 66 percent of respondents prioritizing these technologies for operational optimization, customer engagement, and decision-making insights. As mentioned, agentic AI made a strong debut according to 47 percent of respondents, establishing itself as a core technology focus in its first year of measurement.

Larger companies are leading these AI trends, with 77 percent prioritizing data analytics and 72 percent focusing on generative AI. Smaller companies, often more resource constrained, are leaning more heavily on agentic AI. Fifty-four percent are using or assessing it compared with 40 percent for larger companies, reflecting the power of agentic AI to help companies extend capacity and automate with fewer resources.

Agentic AI engagement is higher among smaller companies (54%) than larger ones (40%).

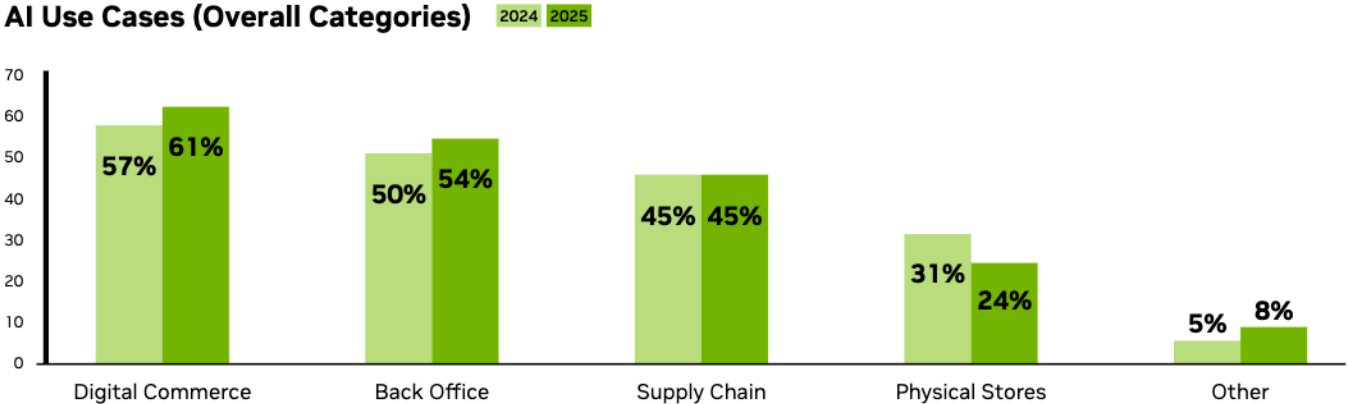
Top 3 AI Areas Being Used or Assessed



The Growth of Customer-Facing and Back-Office Use Cases

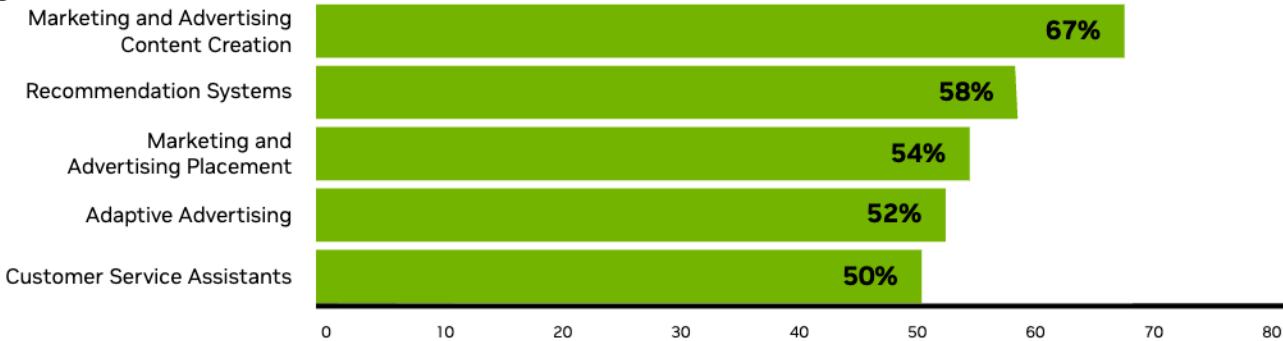
AI adoption continues to rise across major business functions, with digital commerce growing through customer-facing applications, back offices strengthening through analytics and automation, and supply chains holding steady as mature, established sources of operational value. Collectively, these trends point to an industry continuing to invest in both operational enablement and customer-driven innovation—signaling broader confidence in deploying AI where it most directly impacts growth and competitiveness.

AI Use Cases (Overall Categories)

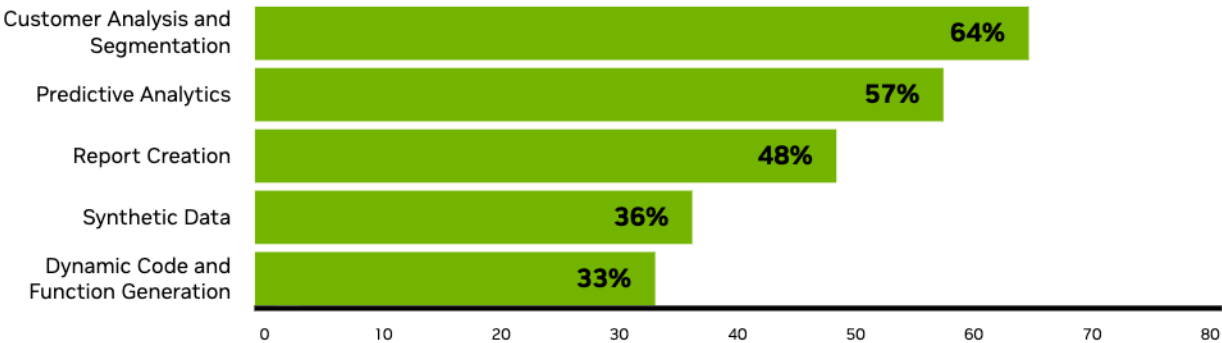


Top 5 Use Cases by Category

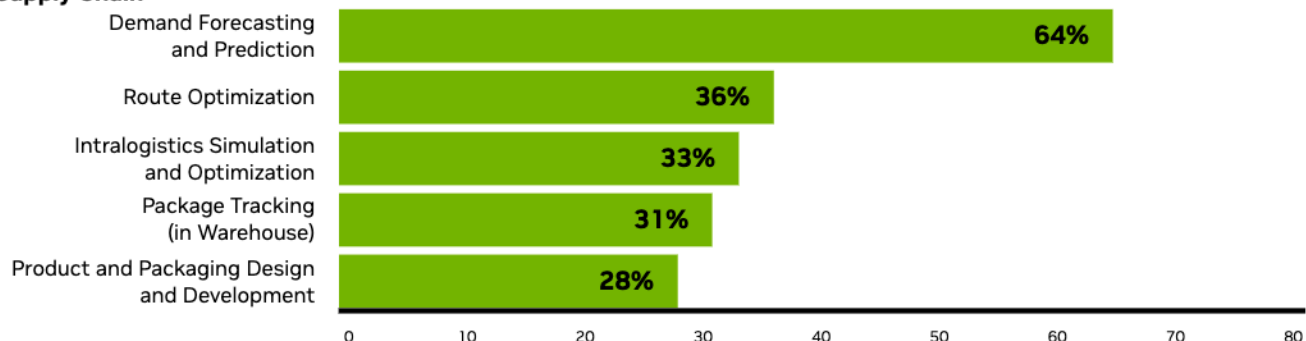
Digital Commerce



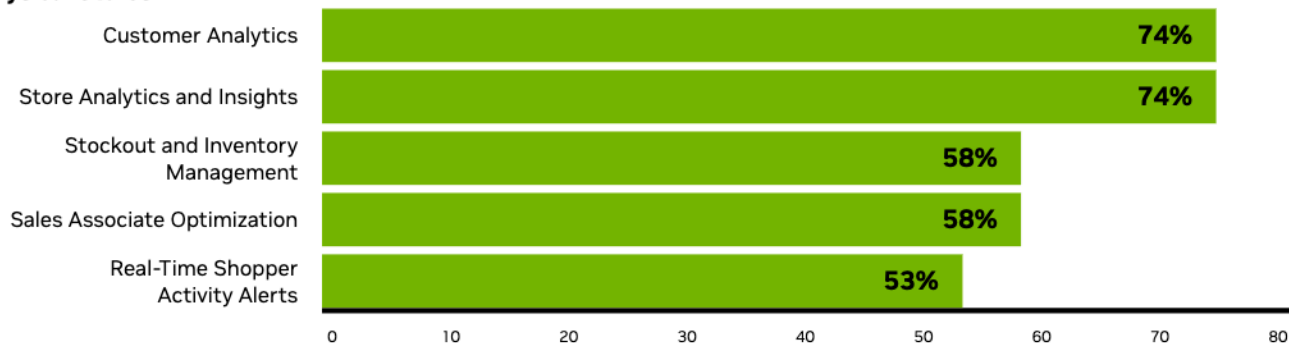
Back Office



Supply Chain



Physical Stores



Customer-Facing Applications: Personalization at Scale

Digital commerce—including ecommerce, marketing, and advertising—continues to lead retail transformation, cited by 61 percent of respondents, up from 57 percent last year. This growth reflects a shift toward AI-driven marketing and personalization, as organizations use intelligent systems to deliver tailored experiences across the customer journey.

61%

are leaning into digital commerce to transform their business.

A major development in this space is agentic commerce, where AI agents shape, curate, and respond to customer intent. Leading product discovery applications include:

- > **Catalog enrichment, at 42 percent**, which involves dynamically enhancing and localizing product information, including text, images, and video
- > **Shopping assistants, at 31 percent**, which guide customers through personalized discovery and purchasing

Physical store applications, while at only 24 percent of the overall focus, demonstrate deep and data-driven adoption where implemented. Both customer analytics and store analytics are deployed by 74 percent of these adopters. This signals targeted investments in understanding and optimizing in-store customer experiences, which can be used to complement digital personalization strategies.

Employee Productivity: Augmenting Human Capabilities

AI is reshaping how employees work across the enterprise. Back-office operations—including marketing operations, finance, product development,

and IT—represent the second-largest area of AI adoption at 54 percent. Organizations are deploying AI to augment human capabilities, driving faster decision-making, greater efficiency, and deeper insights.

Operational Excellence

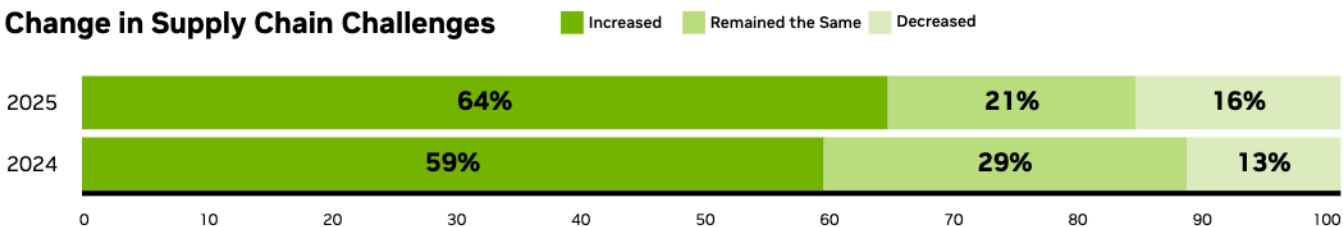
Organizations are increasingly using AI to transform supply chain performance, with 45 percent of respondents citing AI as critical to achieving operational excellence. These initiatives focus on solving complex, high-impact optimization challenges across the end-to-end supply chain, from planning and sourcing to fulfillment and logistics.

Supply Chain Resilience: AI as a Strategic Imperative

Supply chain pressures have surged, with 64 percent of respondents reporting increased challenges year over year. This escalation reflects a convergence of structural shifts: geopolitical instability, labor constraints, evolving consumer expectations for speed and transparency, and regulatory complexity across global operations. The supply chain has evolved from a cost-focused function to a strategic capability that drives customer satisfaction, competitive advantage, and business continuity.

64%
cited increasing supply chain challenges.

Change in Supply Chain Challenges



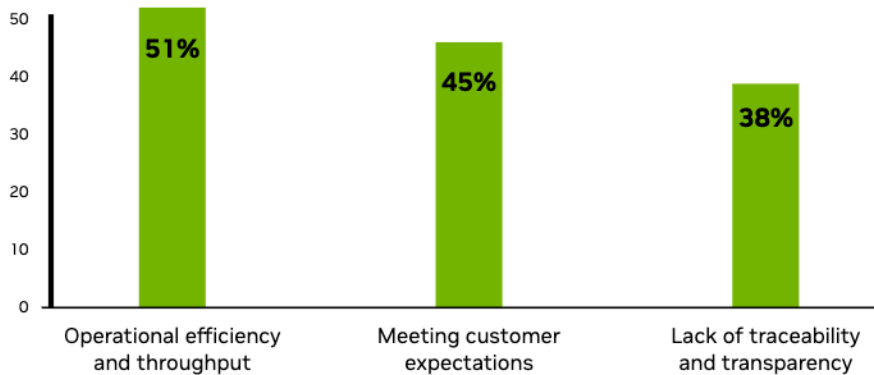
A Multi-Dimensional AI Response

Organizations are deploying AI across multiple supply chain challenges, with each application targeting specific business pressures:

- 1. Operational efficiency and throughput, at 51 percent,** remains the primary focus, reflecting the fundamental need to move goods faster and more cost-effectively. AI-driven automation, predictive maintenance, and process optimization help organizations extract more value from existing infrastructure without proportional capital increases.
- 2. Meeting customer expectations, at 45 percent,** underscores how supply chain performance has become inseparable from customer experiences. AI enables the demand sensing, inventory positioning, and fulfillment orchestration necessary to deliver on promises of fast, accurate, and flexible delivery—capabilities customers increasingly expect as standard.
- 3. Traceability and transparency, at 38 percent,** saw a 15-point increase from last year. This reflects mounting pressure from multiple stakeholders: consumers demanding visibility into product origins and sustainability practices, regulators requiring compliance documentation, and enterprises

needing to manage risk across complex, multi-tier supplier networks. AI-powered track-and-trace systems, supplier monitoring, and provenance verification are evolving from nice-to-have capabilities to business requirements.

Top 3 Supply Chain Challenges Addressed With AI



Physical AI and Robotics: The Start of Next-Gen Supply Chains

Adoption of physical AI and robots may be in its early stages, but the value proposition is becoming increasingly clear. Currently, 17 percent of respondents report using or evaluating physical AI and robotics. Early movers demonstrate that, when integrated thoughtfully, these technologies deliver more than task automation. They enhance flexibility and throughput in response to workforce pressures and rising logistical complexity.

Among those investing in supply chain AI, deployments of physical AI focus on intralogistics simulation and optimization (33 percent), robotic pick and place (23 percent), and smart forklifts and autonomous mobile robots (18 percent).

As pilot initiatives evolve into coordinated systems—where physical AI, digital intelligence, and human labor operate seamlessly together—these technologies are poised to drive the next wave of supply chain transformation and unlock industry-wide value.

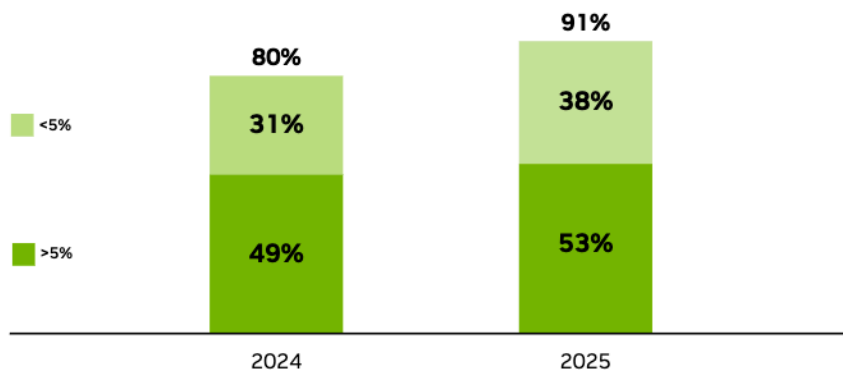
Positive Impact Leads to Sustained Investment for AI in the Supply Chain

Organizations that excel in supply chain AI can build compounding advantages in cost efficiency, customer satisfaction, and operational resilience, making it increasingly difficult for competitors to match. Currently, 91 percent of respondents report that AI has reduced operational costs in the supply chain, up from 80 percent last year.

91%

said AI has reduced operational costs in their supply chain.

AI's Impact on Decreasing Supply Chain Operational Costs



This underscores the technology's growing impact and the rationale for continued investment. Seventy-five percent reported that they'll increase AI spending in the supply chain next year, including 22 percent who said spending will increase by more than 10 percent.

75%

will increase spending on supply chain AI next year.

Shifting Challenges and the Importance of Inference

As generative AI matures, organizations are moving past early hurdles and focusing on execution, which involves balancing data readiness, talent constraints, and emerging technologies.

Data Maturity Progression

Early in the generative AI era (starting in 2022), data was a major obstacle. Enterprises spent years acquiring, organizing, and preparing their data to make it universally accessible for AI-driven problem solving. This year's *NVIDIA State of AI* surveys indicate significant progress: Fewer respondents cite data as a top issue. In retail and CPG, sufficient data for model training dropped as a concern from 27 percent last year to only 13 percent this year, while privacy and sovereignty issues fell from 22 percent to 18 percent.

Only **13%**

said training data was a top challenge this year.

Talent Gap: The Constraining Factor

The AI talent shortage has intensified, rising from 31 percent last year to 46 percent this year, and is now the primary implementation barrier. This increase reflects both the surge in AI adoption and the rise of specialized technologies like agentic AI.

The shortage has real business implications. While 92 percent of executives plan to increase AI budgets, the lack of skilled talent can create a bottleneck where investment appetite exceeds execution capacity. Organizations that address this gap through hiring, strategic partnerships, or upskilling will gain a decisive advantage, translating AI ambitions into operational impact.

A lack of AI talent is this year's top-cited barrier to AI implementation.

The Importance of Inference

AI inference—the process by which a trained model generates predictions and applies learned knowledge to new data—is the engine behind every AI interaction. Every time a model is queried, inference occurs, driving real-time decision-making, recommendations, and operational actions. Optimizing inference is critical to translating AI investments into measurable business value.

Key Considerations for Inference:

> Cost efficiency and total cost of ownership: 41 percent

Production-scale inference costs can quickly erode the margins AI was designed to improve. For retailers processing millions of daily recommendations, personalization requests, or inventory predictions, even fractions of a penny per query accumulate into substantial annual expenses. Organizations that fail to optimize inference risk turning AI from a profit driver into a cost center, especially in high-volume, low-margin operations.

> Latency, accuracy, and throughput: 35 percent

These technical factors directly affect customer experiences and operational effectiveness. Millisecond delays impact conversion rates, limited throughput hampers performance during peak demand, and accuracy issues erode trust, increase support costs, and cascade into inventory imbalances and fulfillment inefficiencies. Models that perform well in testing may falter under production load, creating bottlenecks that limit growth and ROI.

> Model performance and benchmarking: 32 percent

As AI adoption becomes widespread, competitive advantage depends on having superior models. Even small gaps in accuracy or effectiveness can increase costs, reduce sales, and diminish customer lifetime value. Without benchmarking, organizations risk investing in underperforming models, misallocating resources, and missing optimization opportunities.

These factors are interconnected. Optimizing inference delivers compounding benefits—broader deployment, faster optimization cycles, and better outcomes—while failure leaves AI trapped in small-scale pilots, limiting impact and growth. In response, as noted earlier, many organizations are shifting to open, flexible stacks that combine open-source and proprietary models to meet their inference requirements at scale.

Optimizing inference delivers compounding benefits.

Looking Forward

The retail and CPG industry stands at a transformative moment. The maturation of data infrastructure, combined with the rising importance of open-source AI components, the emergence of agentic AI, and the growing momentum of physical AI for supply chains and operations, positions organizations to deliver unprecedented levels of personalization, efficiency, and responsiveness.

The strategic narrative has shifted from experimentation to execution, from back-office automation to customer-facing innovation. Organizations that successfully address the talent gap while scaling agentic AI capabilities will establish competitive advantages in an environment where customer expectations, supply chain complexity, and operational demands continue to intensify.

Investment momentum remains strong, technology foundations are maturing, and business results are validating the strategic bet on AI. The industry is moving beyond asking whether to invest in AI to optimizing how to deploy it most effectively across the value chain—from supply chain prediction to personalized customer engagement.

The next twelve months will be defined by deeper integration of open-source components into production stacks, the operationalization of agentic AI, the continued evolution to specialized AI applications, and the ability to attract and develop the talent necessary to sustain innovation velocity. Organizations that effectively execute these imperatives won't just optimize existing operations. They'll reimagine what's possible in retail and CPG.

The industry is moving beyond asking whether to invest in AI to optimizing how to deploy it.

Methodology

Fielded from August to September 2025, this survey garnered responses from hundreds of respondents, including executives, managers, and practitioners across a range of industry segments in retail and CPG. Segments encompassed apparel and footwear, auto parts and accessories, cleaning and toiletries, convenience stores, cosmetics and beauty, department stores, ecommerce, food and drink, home improvement, home goods, office supplies, pharmacy, specialty and sporting goods, supermarkets and grocers, and warehouse clubs and supercenters. The online survey was sourced from NVIDIA's distribution lists and through social media globally.

Ready to Get Started?

To learn more about how leading retailers are using AI, visit: nvidia.com/retail

